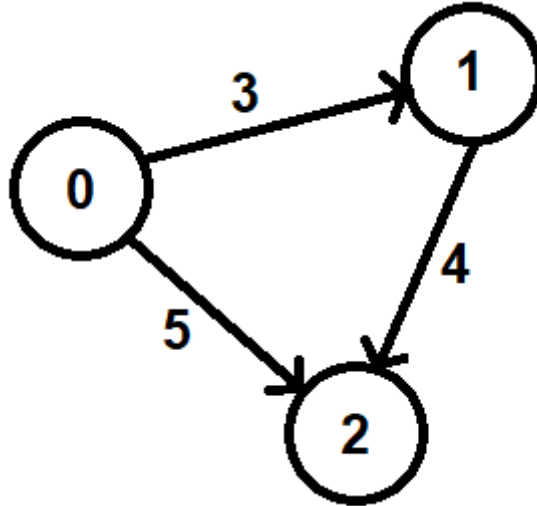


Adjacency Matrix

Adjacency Matrix is a 2D array where rows and columns represent nodes. If there is an edge from node i to node j , then the cell at $[i][j]$ stores the weight of that edge. If there is no edge, that cell stores 0.



The adjacency matrix representation of the above graph is:

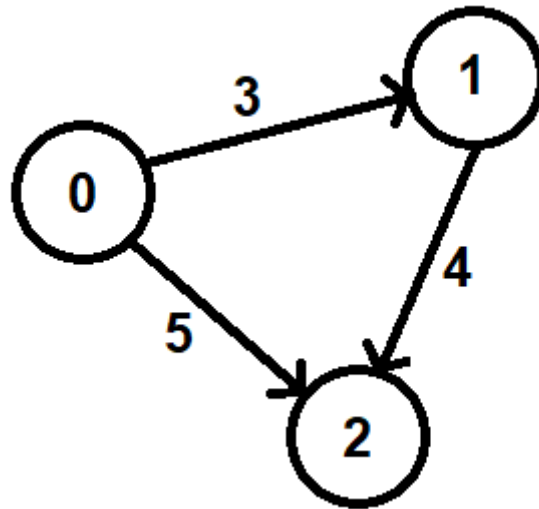
	0	1	2
0	0	3	5
1	0	0	4
2	0	0	0

Coding Idea:

Java	Python
<pre>int[][] graph= new int[n][n]; for (int i = 0; i < m; i++){ graph[u][v] = w; }</pre>	<pre>graph= [] for i in range(n): row = [0] * n graph.append(row) for i in range(m): graph[u][v] = w</pre>

Adjacency List

Adjacency Matrix is a hashmap or dictionary where each key is a node, and its value is a list of all nodes it connects to, usually with weights.



The adjacency list representation of the above graph is:

```
graph = {  
    0: [(1, 3), (2, 5)],  
    1: [(2, 4)],  
    2: []  
}
```

Coding Idea:

Java	Python
<pre>HashMap<Integer, ArrayList<int[]>> adj = new HashMap<>(); for (int i = 0; i < n; i++) { adj.put(i, new ArrayList<>()); } for (int i = 0; i < m; i++) { adj.get(u[i]).add(new int[]{v[i], w[i]}); }</pre>	<pre>graph = {} for i in range(n): graph[i] = [] for i in range(m): adj[u[i]].append((v[i], w[i]))</pre>